01 · Executive Summary

PolicyGuard AI: an intelligent compliance assistant for insurance underwriting document audit and policy renewal TEAM011 · Joydev Pain (2025MB26105) · MBA ZG583 Management of AI Products

The Milestone 1 title ended in "Automated Policy Renewal". I dropped "Automated" because a person always makes the renewal decision.

Problem

Insurers keep a document folder for every policy. Regulators and internal audit expect each folder to follow a set structure, use standard file names, and hold complete, consistent documents. Today, staff check folders by hand, which has four results:

- **Late discovery:** gaps often surface only during a third-party audit or at renewal.
- **Lost analyst time:** hours go to routine checks instead of underwriting judgment.
- **Inconsistent standards:** naming and structure vary across business lines and legacy systems.
- **Weak renewals:** renewals stall, or proceed on incomplete documents.

Target user

Primary: underwriting operations analysts who check policy folders. **Secondary:** compliance and audit teams, renewal teams, underwriting managers and risk officers.

Product concept

An analyst chooses one policy folder. PolicyGuard AI reads each document and identifies its type from content. It then checks structure and naming with plain rules, retrieves the relevant compliance rules through **RAG**, and compares policy numbers, dates and forms across documents. The output is an **audit-readiness score**, findings linked to a rule, quote and confidence band, and a **renewal suggestion**.

A **person decides every finding, gives a reason and signs off**, which creates an audit log and report. The AI never approves a renewal or closes a finding. If analysis fails, it gives no score and no recommendation.

Why AI?

Alternative	Why it falls short
Rule-based file checks	They see names, not content: they found 8 of 30 planted defects in testing
Repository or dashboard	It stores documents but cannot judge whether they are complete or consistent
Checklist or more analysts	Reading stays manual, so cost rises with volume and consistency remains a problem

The work needs someone to read varied documents, recognize one concept under different labels, and link evidence to a rule. AI does that part. Plain rules keep the exact checks, such as date comparisons.

Evidence so far

A working web prototype covers the full loop. I tested it on 24 synthetic folders (95 PDFs, 30 planted defects). The 12 holdout folders used different labels and date formats, and were run once with no tuning afterward.

Holdout result (12 folders, 16 defects)	PolicyGuard AI	File-name-only baseline
Defects found	**16 of 16**	4 of 16

Holdout result (12 folders, 16 defects)	PolicyGuard AI	File-name-only baseline
False alarms	**1**	2
Critical defects missed (of 3)	**0**	2
Renewal suggestions as expected	**12 of 12**	Not applicable

- **It fails safely:** in a real quota outage, 11 folders were marked Not Scored, and none received a false pass.
- **This is not real-world accuracy:** the data is synthetic and simple, the test set is small, and one finding changed between runs.
- **Time savings carry the business case:** on planning assumptions (a US insurer, onsite analysts at US$150 an hour), time savings alone pay back the US$360,000 investment in about 8–10 months. Avoided audit remediation is upside, not relied on.

Recommendation: PILOT

Run a **12-week controlled pilot in commercial property with 6–8 analysts**, starting in shadow mode, on an enterprise AI agreement and the insurer's real rulebook.

- **Enter the pilot only if** offline tests on a 300-folder expert-labeled set show at least 95% critical-defect recall, no fabricated evidence in 100 audited findings, and at least 95% repeatability.
- **Scale only if** the pilot shows at least 40% less analyst time and at least 70% of reviewers rate the tool useful, with a rejection rate of 10–40%.
- **Stop if** critical misses cannot be controlled, reviewers reject more than 40% of findings, or privacy approval is refused.